Certified Lattice Heat Expansions and Directional Capacity Errors for the First Correction to Bloch's Law

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Abstract

We study the precise interfaces in the source manuscript *The first lattice correction to Bloch's law*, keeping fixed spin, slot normalization, nonreversible period kernels, and the ordered thermodynamic, field, and low-temperature limits. An exact change of variable in the lattice heat integral gives signed, all-order remainder bounds with explicit constants. These yield uniform anisotropic Bose-sum expansions, including fugacity one, and certified tails after subtraction of finitely many asymptotic terms. At the source's capacity interface we retain separate forward and reverse residuals, solve the resulting quadratic energy inequality exactly, and identify the optimal homogeneous energy norm on the actual zero-boundary test space when its additional range condition holds. A finite error ledger connects these refinements without inferring an interacting third coefficient from a more accurate ideal density. We correct one slot/site identification in the pressure module; the source's later normalization and limit order are retained. Classical heat coefficients and duality are not claimed as new principles. Contribution assessment: C3/5, explicit uniform lattice-sum remainder certificates and a sharper directional capacity interface provide useful local refinements without independently establishing the global interacting asymptotic.

1 Physical normalization and the scope of the question

Let $S \in \{\frac{1}{2}, 1, \frac{3}{2}, \dots\}$ be fixed and set $\ell = 2S$. On a cube $\Lambda \subset \mathbb{Z}^3$ the source uses

$$H_\Lambda = - \sum_{\{x,y\}\subset\Lambda:\, |x-y|=1} \mathbf{S}_x \cdot \mathbf{S}_y, \qquad M_\Lambda = \sum_{x\in\Lambda} S_x^3.$$

Each unordered edge is counted once, with free spatial boundary before the auxiliary boundary pinning. The infinite-volume pressure is

$$p(\beta, h) = \lim_{\Lambda\uparrow\mathbb{Z}^3} \frac{\log \operatorname{Tr} e^{-\beta(H_\Lambda - hM_\Lambda)}}{\beta|\Lambda|}, \qquad m(\beta) = \partial_h^+ p(\beta, 0).$$

The source claims, for fixed S,

$$S - m(\beta) = n_{\mathrm{sw}}(\beta) + o((\beta S)^{-5/2}), \qquad n_{\mathrm{sw}}(\beta) = \int_{[-\pi,\pi]^3} \frac{dk/(2\pi)^3}{e^{2\beta S\sum_j(1-\cos k_j)} - 1}. \tag{1}$$

The proof of this interacting comparison is an imported global claim in this report. Reading its proof does not turn the present local analysis into an independent verification of all loop, ordering, geometry, and uniform pinning inputs.

Write $t = \beta S$. For the ideal quantity, the source already proves

$$n_{\mathrm{sw}}(\beta) = \frac{\zeta(3/2)}{8\pi^{3/2}} t^{-3/2} + \frac{3\zeta(5/2)}{128\pi^{3/2}} t^{-5/2} + O(t^{-7/2}). \tag{2}$$

Neither these coefficients nor the existence of an ideal expansion are new here. Our additions are explicit signed remainders at every fixed order, their uniform summation and finite-tail use, and an independent refinement of the nonreversible comparison error.

1.1 The slot/site correction

In the full-space free model, a slot walk has rate $1/2$ to each neighboring slot. Its site projection has jump rate S per neighboring site. Let $p_s(x)$ be that site's heat kernel. The seam averages the ℓ slots at a site. With J this averaging projection,

$$U_0 = P_\beta J, \qquad U_0^n((x,\sigma),(y,\tau)) = \ell^{-1} p_{n\beta}(y-x).$$

Consequently the source profile at its distinguished site is

$$v(x) = \ell^{-1} \sum_{n \geq 1} p_{n\beta}(0) = \ell^{-1} n_{\mathrm{sw}}(\beta), \qquad c = (1 + v(x))^{-1}. \tag{3}$$

Tonelli justifies the nonnegative heat expansion of the Bose denominator; the integral at zero momentum is finite in dimension three.

The sentence after this slot sum in `08-pressure.tex`, lines 56–57, identifies it with n_{sw}. It must instead identify it with n_{sw}/ℓ. This is a local transcription error when $\ell \neq 1$, not a change to the source's final coefficient: `04-profiles.tex` explicitly states $\ell v = n_{\mathrm{sw}}$; the later pressure calculation uses $r = \ell^{-1} n_{\mathrm{sw}} + o(\beta^{-5/2})$ and sums the ℓ slots. We retain those correct statements. The cached original is not edited.

2 A signed, explicit heat-kernel remainder

For $t > 0$, define

$$I(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-2t(1-\cos k)}\, dk = e^{-2t} I_0(2t), \qquad K(t) = \sqrt{4\pi t}\, I(t). \tag{4}$$

The Bessel identity is classical [2]. We will not use a formal Bessel asymptotic as a remainder estimate. Put

$$b_j = \frac{\binom{2j}{j}}{4^j}, \qquad a_j = \frac{((2j)!)^2}{64^j (j!)^3}, \qquad P_m(t) = \sum_{j=0}^{m} a_j t^{-j} \quad (m \geq 0).$$

In particular $a_0 = 1, a_1 = 1/16, a_2 = 9/512, a_3 = 75/8192$. All cutoffs and constants below refer to the actual Brillouin-zone integral, not a polynomial dispersion on an unbounded momentum space.

Theorem 2.1 (Two-sided finite-order enclosure)**.** *For every integer $m \geq 0$ and every $t \geq 1$,*

$$-T_m(t) \leq K(t) - P_m(t) \leq 2a_{m+1}t^{-m-1} + \sqrt{\pi t}\, e^{-2t}, \tag{5}$$

where

$$T_m(t) = \frac{1}{\sqrt{\pi}} \sum_{j=0}^{m} \frac{b_j}{4^j t^j} \Gamma(j + \tfrac{1}{2}, 2t) \leq \sqrt{2} e^{-t} \sum_{j=0}^{m} 2^j a_j t^{-j}. \tag{6}$$

Here $\Gamma(s,x) = \int_x^\infty u^{s-1}e^{-u}\,du$. In particular $|K(t) - P_m(t)| \leq C_m t^{-m-1}$, with the explicit choice

$$U(b,c) = \sup_{t \geq 1} t^b e^{-ct} = \begin{cases} e^{-c}, & b \leq c, \\ (b/(ce))^b, & b > c, \end{cases} \tag{7}$$

$$C_m = \max\left\{ 2a_{m+1} + \sqrt{\pi}\, U(m + \tfrac{3}{2}, 2),\ \sqrt{2} \sum_{j=0}^{m} 2^j a_j U(m+1-j, 1) \right\}. \tag{8}$$

These are finite-order bounds; no convergence of the infinite asymptotic series is asserted.

Proof. The monotone substitution $x = 2\sin(k/2)$ gives the exact expression

$$I(t) = \frac{1}{2\pi} \int_{-2}^{2} e^{-tx^2} (1 - x^2/4)^{-1/2}\, dx. \tag{9}$$

The integrable endpoint singularity is essential; it cannot simply be discarded in a uniform Taylor argument. Since $b_{j+1}/b_j = (2j+1)/(2j+2) \leq 1$, for $0 \leq z \leq 1/2$

$$0 \leq (1-z)^{-1/2} - \sum_{j=0}^{m} b_j z^j \leq b_{m+1} \frac{z^{m+1}}{1-z} \leq 2b_{m+1} z^{m+1}.$$

On $|x| \leq \sqrt{2}$, insert $z = x^2/4$. Integration against the Gaussian on the whole line bounds the normalized remainder by $2a_{m+1}t^{-m-1}$, because

$$\frac{\sqrt{t}}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-tx^2} \frac{b_j x^{2j}}{4^j}\, dx = a_j t^{-j}.$$

The arcsine measure in $\sqrt{2} < |x| < 2$ has mass $1/2$; hence its normalized contribution is at most $\sqrt{\pi t} e^{-2t}$. The part of the Gaussian polynomial integral outside $[-\sqrt{2}, \sqrt{2}]$ is precisely $T_m(t)$. The central remainder and the true outer integral are nonnegative, giving both sides of (5).

On that outside region, $e^{-tx^2} \leq e^{-t} e^{-tx^2/2}$. Integrating each monomial on the whole line with variance doubled proves (6). Finally multiply the positive and negative bounds by t^{m+1}. Elementary differentiation gives (7) and then (8). □

Remark 2.2. The local moment expansion has been performed in an exact coordinate, with the Jacobian retained and the complementary region bounded. This avoids using a small-momentum Taylor polynomial beyond its domain of validity. It also makes the sign of each discarded contribution visible. The constants grow with m; increasing the order indefinitely at a fixed t is not justified.

3 Uniform anisotropic Bose sums and accelerated tails

Fix $\nu = (\nu_1, \nu_2, \nu_3) \in (0, \infty)^3$, and $0 \le z \le 1$. The following is an ideal lattice model, not an assertion about an anisotropic interacting ferromagnet:

$$N_\nu(t, z) = \int_{[-\pi,\pi]^3} \frac{z\, dk/(2\pi)^3}{e^{2t\sum_j \nu_j(1-\cos k_j)} - z} = \sum_{n\ge 1} z^n \prod_{j=1}^{3} I(\nu_j n t). \tag{10}$$

At $z = 0$ both sides are zero. At $z = 1$ the integrable zero-momentum singularity is understood in the Lebesgue sense. Set

$$A_\nu = (4\pi)^{-3/2}(\nu_1\nu_2\nu_3)^{-1/2}, \quad \prod_{j=1}^{3} P_m(\nu_j t) = \sum_{h=0}^{3m} c_h(\nu) t^{-h},$$

and define the following explicit, nonnegative constants:

$$B_j = \sum_{h=0}^{m} a_h \nu_j^{-h}, \qquad R_j = C_m \nu_j^{-m-1}, \qquad D_j = B_j + R_j,$$
$$E_{m,\nu} = \sum_{j=1}^{3} R_j \prod_{i<j} D_i \prod_{i>j} B_i + \sum_{h=m+1}^{3m} c_h(\nu).$$

An empty last sum is zero. The order of the telescoping products is fixed here; other orders can give different valid numerical constants.

Theorem 3.1 (Summable remainder including fugacity one)**.** *For $t \ge t_\nu := \max(1, \nu_1^{-1}, \nu_2^{-1}, \nu_3^{-1})$,*

$$\left| N_\nu(t, z) - A_\nu t^{-3/2} \sum_{h=0}^{m} c_h(\nu) t^{-h} \operatorname{Li}_{h+3/2}(z) \right| \le A_\nu E_{m,\nu} t^{-m-5/2} \operatorname{Li}_{m+5/2}(z). \tag{11}$$

The estimate is simultaneous for $0 \le z \le 1$; constants depend on the fixed positive anisotropy and fixed order, not on z.

Proof. For every $u \ge t_\nu$, $|K(\nu_j u) - P_m(\nu_j u)| \le R_j u^{-m-1}$, $P_m(\nu_j u) \le B_j$, and $|K(\nu_j u)| \le D_j$. Telescoping the three products and then retaining only powers through m gives

$$\left| \prod_j K(\nu_j u) - \sum_{h=0}^{m} c_h(\nu) u^{-h} \right| \le E_{m,\nu} u^{-m-1}.$$

In the polynomial tail we used $u \ge 1$ and positive coefficients. Multiply by $A_\nu u^{-3/2}$, put $u = nt$, and sum with weights z^n. The dominating series has exponent $m + 5/2 > 1$, even at $z = 1$. The original heat sum follows from nonnegative Tonelli. Thus no exchange of an unbounded, nonuniform asymptotic remainder with the Bose series is involved. □

Corollary 3.2 (Third ideal coefficient)**.** *For $m = 2$,*

$$c_0 = 1, \quad c_1 = \frac{1}{16}\sum_j \nu_j^{-1}, \quad c_2 = \frac{9}{512}\sum_j \nu_j^{-2} + \frac{1}{256}\sum_{j<k} (\nu_j\nu_k)^{-1}.$$

For $\nu = (1,1,1)$, $c_1 = 3/16, c_2 = 33/512$, and

$$N_{\mathbf{1}}(t,1) = \frac{\zeta(3/2)}{8\pi^{3/2}} t^{-3/2} + \frac{3\zeta(5/2)}{128\pi^{3/2}} t^{-5/2} + \frac{33\zeta(7/2)}{4096\pi^{3/2}} t^{-7/2} + \mathcal{R}(t), \qquad |\mathcal{R}(t)| \le A_{\mathbf{1}} E_{2,\mathbf{1}} \zeta(9/2) t^{-9/2}. \tag{12}$$

The coefficient is classical lattice spin-wave information, not a new interacting correction. An $o(t^{-5/2})$ comparison error in (1) can be larger than $t^{-7/2}$.

Example 3.3. For $\nu = (1,2,4)$,

$$A_\nu = \frac{1}{16\sqrt{2}\,\pi^{3/2}}, \qquad c_1 = \frac{7}{64}, \qquad c_2 = \frac{217}{8192}.$$

The theorem applies for $t \ge 1$, including $z = 1$. These exact coefficients retain both repeated-axis and mixed-axis quartic effects. Setting a weight to zero is not a limiting operation covered by the constant: the zero-fugacity-gap integral in effective dimension two diverges. Uniformity as $\min_j \nu_j \downarrow 0$ is not claimed.

Corollary 3.4 (Finite residual-sum certificate)**.** *Let $B_n = \prod_j I(\nu_j n t)$, and let*

$$H_n = A_\nu (nt)^{-3/2} \sum_{h=0}^{m} c_h(\nu) (nt)^{-h}.$$

For an integer $N \ge 1$, evaluate only the first N residuals:

$$\mathcal{A}_N = A_\nu t^{-3/2} \sum_{h=0}^{m} c_h(\nu) t^{-h} \operatorname{Li}_{h+3/2}(z) + \sum_{n=1}^{N} z^n (B_n - H_n).$$

Then

$$|N_\nu(t,z) - \mathcal{A}_N| \le \frac{A_\nu E_{m,\nu}}{m+3/2}\, t^{-m-5/2} N^{-m-3/2}. \tag{13}$$

If the finite values B_n have certified errors ϵ_n, add $\sum_{n \le N} z^n \epsilon_n$; errors in the displayed polylogarithms and arithmetic must also be included.

Proof. Use the pointwise bound in the proof of Theorem 3.1 and $\sum_{n>N} n^{-m-5/2} \le \int_N^\infty x^{-m-5/2}\, dx$. The finite error term is the ordinary triangle inequality. □

Direct summation only gives the corresponding $N^{-1/2}$ absolute tail bound. At $m = 2$, residual subtraction gives $N^{-7/2}$. This is a certified tail improvement, not a measured speedup, a floating-point stability claim, or permission to omit special-function evaluation errors. Cancellation in $B_n - H_n$ can require extra precision.

4 The two directional residuals of a killed period

The heat expansion diagonalizes the free translation-invariant operator. It does not diagonalize a time-ordered period in a random hole geometry. We now work directly with the latter finite operator and the counting inner product, as required by the source.

Let H be a finite coordinate set, $K \subset H$, $i \notin K$. Let U be entrywise nonnegative, with every row and column sum at most one and zero rows and columns on K. Kernels act on columns; thus

U_{ba} is the transition mass from a to b. Missing mass is killed. Put $A = I - U$. The inequality $\|U\|_{2\to 2} \le 1$ follows either by weighted Cauchy–Schwarz in each row and summing, or from the row/column norm bound. Thus

$$e_z = \langle z, Az\rangle = \langle z, S_0 z\rangle \ge 0, \qquad S_0 = (A + A^*)/2.$$

No symmetry of U is assumed.

Let $f(a)$ be the probability of ever hitting i before killing, with $f(i) = 1, f|_K = 0$, and let q be the probability of a return to i after at least one step starting at i. The minimal hitting solution satisfies

$$A^* f = (1 - q)\eta_i, \qquad 0 \le f \le 1. \tag{14}$$

This statement remains true with a closed class avoiding i; almost-sure killing is not needed. The minimal solution is zero on such a class, even though arbitrary harmonic solutions need not be.

Choose a real profile h with $h_i = 1, h|_K = 0, \|h\|_\infty \le M$ and a real number c. Define

$$\mu_+ = Uh - h + c\eta_i, \quad \mu_- = U^* h - h + c\eta_i, \quad Z = \{z \in \mathbb{R}^H : z_i = 0,\ z|_K = 0\}.$$

In the application h is the spatially cut off all-turn free profile, and $0 \le h \le 1$. The two errors are genuinely different because the reverse period inverts the ordering, seam and transports.

Lemma 4.1 (Exact identities on the actual test space)**.** *For $y = f - h \in Z$,*

$$e_y = \langle y, \mu_-\rangle, \qquad q - (1 - c) = \langle h, \mu_+\rangle + \langle y, \mu_+\rangle. \tag{15}$$

Proof. Subtract A^*h from (14) and pair with y. The coefficient of η_i disappears since $y_i = 0$; $\langle y, A^* y\rangle = e_y$ for real vectors. For the second identity,

$$\langle f, \mu_+\rangle = -\langle A^* f, h\rangle + cf_i = -(1 - q) + c.$$

Insert $f = h + y$. These are the source's two-residual identities, not new capacity principles. □

Theorem 4.2 (Separate directional budgets)**.** *Suppose $X_\pm, \epsilon_\pm \ge 0$, and for each $z \in Z$ with $\|z\|_\infty \le 1 + M$,*

$$|\langle z, \mu_\pm\rangle| \le X_\pm \sqrt{e_z} + \epsilon_\pm. \tag{16}$$

Define $\tau(X, \epsilon) = (X + \sqrt{X^2 + 4\epsilon})/2$. Then

$$|q - (1 - c)| \le |\langle h, \mu_+\rangle| + X_+ \tau(X_-, \epsilon_-) + \epsilon_+. \tag{17}$$

For $\epsilon_- = 0$, the mixed term is exactly $X_+ X_-$. If $X_\pm \le X$ and $\epsilon_\pm \le \epsilon$, the right side is no larger than the source's valid $|\langle h, \mu_+\rangle| + 2X^2 + 2\epsilon$.

Proof. The vector y has the required sup norm. By Lemma 4.1, $e_y \le X_- \sqrt{e_y} + \epsilon_-$. Writing $u = \sqrt{e_y} \ge 0$ gives $u \le \tau(X_-, \epsilon_-)$. Use the other identity and the plus estimate. For the last comparison, $\tau(X, \epsilon) \le X + \sqrt{\epsilon}$, and $X\sqrt{\epsilon} \le (X^2 + \epsilon)/2$; hence $X\tau + \epsilon \le \frac{3}{2}(X^2 + \epsilon) \le 2X^2 + 2\epsilon$. □

Corollary 4.3 (Random good-set error, without independence)**.** *Let the preceding objects depend measurably on a picture, let $0 \le c \le 1$ be deterministic, and suppose the hypotheses hold on an event G, with deterministic $\epsilon_\pm$. Then*

$$\begin{aligned}\mathbb{E}|q - (1 - c)| \le{}& \mathbb{E}\mathbf{1}_G |\langle h, \mu_+\rangle| + \sqrt{\mathbb{E}\mathbf{1}_G X_+^2\, \mathbb{E}\mathbf{1}_G X_-^2} \\ &+ \sqrt{\epsilon_-\, \mathbb{E}\mathbf{1}_G X_+^2} + \epsilon_+ + \mathbb{P}(G^c).\end{aligned} \tag{18}$$

Proof. Use $\tau(X_-, \epsilon_-) \le X_- + \sqrt{\epsilon_-}$, then Cauchy–Schwarz on the good event. On its complement both q and $1 - c$ lie in $[0, 1]$. No independence of the residuals or the event is required. □

The exact expected product in Theorem 4.2 may be retained if more informative than its Cauchy–Schwarz upper bound. The source has $\mathbb{E}\mathbf{1}_G(X_+^2 + X_-^2) = o(\beta^{-5/2})$ and a rapidly small additive error for bounded tests. Thus this sharpening preserves its downstream order while retaining asymmetric information and better finite constants. It does not improve the source's probability estimates.

4.1 When a homogeneous dual norm really is available

Proposition 4.4 (Restricted energy norm and null classes)**.** *Compress S_0 to Z, obtaining $S_Z \ge 0$, and let $b_\pm$ be the coordinate restrictions of $\mu_\pm$ to Z. A bound*

$$|\langle z, b_\pm \rangle| \le X_\pm \sqrt{\langle z, S_Z z \rangle} \quad \textit{for all } z \in Z$$

exists with finite $X_\pm$ if and only if $b_\pm \in \operatorname{Ran} S_Z$. The least squared constant is

$$X_\pm^2 = b_\pm^* S_Z^\dagger b_\pm, \tag{19}$$

where $S_Z^\dagger$ is the Moore–Penrose inverse.

Proof. If $z \in \ker S_Z$, the bound forces $\langle z, b_\pm \rangle = 0$. In finite dimension this is equivalent to membership in $\operatorname{Ran} S_Z$. Diagonalize the real symmetric S_Z; on its positive eigenspaces Cauchy–Schwarz gives (19). Taking $z = S_Z^\dagger b_\pm$, or a nonzero multiple, attains the ratio when $b_\pm \ne 0$. The zero case is immediate. □

This is classical energy duality, here restricted to the source's *actual* test space. Deleting only K instead of also deleting i can impose a needless norm cost. Conversely a pseudoinverse alone does not check the range condition: for $S_Z = 0, b = 1$, its proposed quadratic value is zero although the asserted inequality is false. Nor does (16), with bounded tests and nonzero $\epsilon_\pm$, imply this homogeneous all-vector bound. That additional premise must be proved separately in any physical use.

Example 4.5 (Exact asymmetric finite period). Take $K = \varnothing$, $i = 1$, and

$$U = \begin{pmatrix} 0 & 1/2 \\ 3/4 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 \\ 1/4 \end{pmatrix}, \qquad c = 1.$$

The matrix is doubly substochastic but not symmetric. Here $f = (1, 1/2)^*$, $q = 3/8$, $y = (0, 1/4)^*$, and

$$S_Z = (1), \quad b_- = 1/4, \quad b_+ = 1/2, \quad \langle h, \mu_+ \rangle = 1/4.$$

The directional bound is attained: $q = 1/4 + (1/4)(1/2) = 3/8$. The source's coarser common-X estimate with $X = 1/2$ gives $3/4$. Adjoining a disconnected coordinate with $U_{33} = 1, h_3 = 0, f_3 = 0$ makes S_Z singular, yet the compatible residuals and the same bound remain valid. This is a finite illustration of the formulas, not a sampled proof of the random geometric estimates.

5 A finite ledger up to the pressure derivative

We make explicit the boundary between these local results and the uniform interacting input. In the source's exposed loop picture, let q_i be a return probability and $r = \mathbb{E}_{\mathbb{P}_P} q_i$. The exact pin identity is

$$d_E(i) = \frac{r}{1+r}. \tag{20}$$

The expectation is under the source's exposed-picture law; it is not the expectation of $q_i/(1+q_i)$. The source constructs $h = c\chi(\eta_i + v)$, with $c = (1+v(x))^{-1}$, and controls the scalar and both energy residuals uniformly over the relevant cubes and all finite deterministic pins outside $B_{50R_*}(x)$. These global controls are imported.

Proposition 5.1 (Conditional physical error budget)**.** *Suppose, for every such cube and pin set, a deterministic bound $\mathbb{E}|q_i - (1-c)| \le \mathcal{B}$ is established, for example by Theorem 4.2 or Corollary 4.3. With $v = v(x) = n_{\mathrm{sw}}/\ell$,*

$$|d_E(i) - v| \le \mathcal{B} + \frac{2v^2}{1+2v}. \tag{21}$$

If the source's boundary-pressure comparison and field-pinning identities hold with this same uniform budget, then

$$|S - m(\beta) - n_{\mathrm{sw}}(\beta)| \le \ell\mathcal{B} + \frac{2\ell v^2}{1+2v}. \tag{22}$$

One may add the ideal remainder in Theorem 3.1, at $\nu = \mathbf{1}, z = 1$, to compare with any of its finite ideal expansions.

Proof. Put $r_0 = 1 - c = v/(1+v)$. Then $|r - r_0| \le \mathcal{B}$, and the function $r/(1+r)$ is 1-Lipschitz on $[0,\infty)$. Furthermore $r_0/(1+r_0) = v/(1+2v)$. The triangle inequality gives (21).

For clarity, the remaining transfer is conditional on the indicated source identities, not a consequence of a finite matrix estimate. The source's joint spin/pin construction uses $b_\beta(h) = 1 - e^{-\beta h}$. Its marginal pin law is a posterior, not independent Bernoulli. Nevertheless conditional on the colors, only up slots are selected independently; therefore each slot is selected with probability at most $b_\beta(h)$. A union bound on the protected region gives the source error $C_{S,\beta} b_\beta(h)$, at fixed β, while the fraction of sites without the needed interior neighborhood tends to zero as the volume grows. Here $C_{S,\beta}$ is finite and volume-independent; no uniform linear growth bound in β is required.

Summing (21) over the ℓ slots gives, for the boundary-pinned finite-volume pressure derivative, an error bounded by the right side of (22), plus $C_S(1-\rho_\Lambda) + C_{S,\beta} b_\beta(h)$. Integrate this bound from 0 to $h > 0$ and divide by h. First take $\Lambda \uparrow \mathbb{Z}^3$; the surface pressure error and $1 - \rho_\Lambda$ vanish. Next let $h \downarrow 0$, with β fixed. Convexity identifies the right derivative and the last field error vanishes. This proves (22) under the stated global premises. □

Since $v = O(\beta^{-3/2})$, the explicit rational term is $O(\beta^{-3}) = o(\beta^{-5/2})$, for fixed S. Its appearance in this *bound* does not identify a physical β^{-3} coefficient. Similarly the known ideal $t^{-7/2}$ coefficient is not determined for $S - m$ by an $o(t^{-5/2})$ remainder. For example a hypothetical comparison error $t^{-11/4}$ satisfies the latter requirement but dominates the ideal third term. Free-energy convergence by itself also does not authorize differentiating in h at zero.

6 Source architecture and retained obligations

All 14 cached public text files were read in full in bounded literal displays: 2994 lines, including 2732 lines in the nine scientific section modules. The two figures, main file, bibliography and

README were also read. The dependency map below separates that reading from independent proof certification.

Module	Scientific role and boundary
Introduction	Fixed-S coefficient, pressure derivative and comparison claim; ordering and loop inputs include companion results not audited here.
Loops	Fresh randomness on every turn of a repeated exposure, pin identity, counting-space contractions, forward columns and reversed transports. These physical laws are preserved, not replaced by a free heat kernel.
Geometry	Regularity, multipoint sparsity, surgery, local paths, Nash bounds and displacement. The good event has $\mathbb{P}(G^c) \le \beta^{-2.91}+$ rapidly small terms. The isolated-path domination is a joint-law inequality, not a conditional independence assertion.
Capacity	Two residual identities, bounded-test comparison and uniform deterministic-pin proposition. Theorem 4.2 sharpens this finite step only.
Profiles	All-turn free v, $\ell v = n_{\mathrm{sw}}$, cutoff h, distinct forward and reverse seam placement, derivative majorants. Noncompact tails and cutoff annuli are indispensable.
Sources	Continuous and atomic residuals, signed isolated-block cancellation and time-ordered Duhamel pairing. Atomic self-pairs are omitted as prescribed; a seam is not counted twice.
Scalar	Expected scalar residual $o(\beta^{-5/2})$, distinct-site versus diagonal occupation contributions, dyadic time gaps. Different times are handled without an independence assumption.
Energy	Measurable, test-independent $X_{\pm}$, flows and local paths, next-block averaging and a final copied period. The copy contributes an energy correction and is not an independent random sample.
Pressure	Good/bad averaging, nonlinear pin conversion, posterior field selection, volume then field then temperature limits, and the classical quartic ideal expansion. One slot/site sentence is corrected in Section 1.1.

For reference the source uses

$$\delta = 1/200, \quad L = \beta^{\delta}, \quad D_* = \sqrt{\beta}, \quad R = \beta^{3/2+\delta/4}, \quad R_* = \lceil \beta^{5/3} \rceil.$$

Fixed logarithmic powers are absorbed by strict power gains. The regularity, signed source cancellation and flow estimates must be simultaneous with their actual cube and pin uniformity. A small residual for a single finite picture is insufficient. The scalar argument contains, among other contributions, the power $\beta^{-5/2-\delta/4}$; the energy argument likewise retains this strict gain. We do not promote these displayed intermediate powers to an effective global numerical rate with known constants. Several preliminary geometry and ordering inputs come from companion manuscripts. Their proofs were not independently reconstructed here.

In particular the complete interacting assertion (1) requires the exposed-picture law, surgery and multipoint domination, geometry uniformity, both scalar and energy estimates, and the field-pressure passage simultaneously. The present report supplies selected local deductions and a careful normalization repair, not an independent global audit of that chain.

7 Original Ram derivations and what they do not imply

Two confidential original chapter bodies were freshly read: the 446-line *Local moments* chapter and the 373-line *Ram–CS* chapter of volume 1, revision 2 [6, 7]. Their subordinate included files were not read, and no assembled-book or whole-corpus reading is claimed.

The local-moment proof expands along actual line segments and bounds the integrated Taylor remainder by a moment and a derivative majorant. It distinguishes output-centered and source-centered expansion, retains coordinate Jacobians, and warns that polynomial exactness is not an inverse or closure theorem. Theorem 2.1 follows that discipline in a particularly concrete setting: the arcsine coordinate makes the phase exactly quadratic, its Jacobian supplies positive moments, and the complement of the Taylor region is estimated separately. The moment identities by themselves would not control the singular endpoints or the infinitely many Bose turns.

The Ram–CS proof distinguishes an absolutely integrable Fourier representation from an L^2 Plancherel identity; exchanging integrals or differentiating requires the appropriate majorants. It also distinguishes constant convolution symbols from variable-kernel operators and notes that a formal reciprocal need not be a bounded inverse. Here Fourier diagonalization applies to U_0 only. The exposed geometry changes with time and carries ordered seam and transport maps, so its capacity must be handled by the two physical residuals. Replacing that problem by the multiplier of the free model would erase the main source obligation.

The concrete synthesis is therefore not a generic finite-moment replacement for the loop process: it is a globally dominated ideal sum, a restricted counting-energy norm when available, and an error ledger that keeps them separate. The source already supplies the global all-turn geometry; neither original chapter supplies it.

8 Literature, advantages and remaining work

The public primary-source refresh was made on October 9, 2026, using public titles and URLs only. DLMF supplies the classical integral and asymptotic Bessel formulas [2]; the explicit signed remainder in this report was proved directly rather than inferred from a formal asymptotic symbol. Correggi, Giuliani and Seiringer [3] rigorously treat the leading low-temperature free energy at fixed spin. That result is not by itself the field-derivative comparison at the precision in (1). Hofmann's effective-Lagrangian treatment [4] discusses lattice-sensitive low-temperature coefficients and interactions. Its stated framework must not be silently substituted for the source's uniform pinned-period probability estimates. Nonreversible capacity principles have a classical literature [5]; the improvement here is retaining the source's two directional budgets and exact quadratic root, not inventing energy duality. The original proofs of these external papers were not independently audited for this report.

What improves. Equations (5)–(8) turn a fixed-order ideal asymptotic into an explicit enclosure. The anisotropic and fugacity parameters remain in the constants and in the summable dominator. Corollary 3.4 makes a concrete finite-tail improvement. Theorem 4.2 does not discard directional imbalance or replace a solvable quadratic inequality by two applications of Young's inequality. The restricted energy formula checks zero-energy classes rather than assuming invertibility.

What costs remain. The heat constants are deliberately safe and can be large. An asymptotic subtraction can increase roundoff sensitivity; it needs certified special-function values if used numerically. The anisotropic constants deteriorate near a vanishing weight. Exact energy dual

norms may require solving a large singular system, and a global homogeneous bound is stronger than the source's bounded-test estimate with an additive error. The product improvement need not change the dominant bad-event or scalar-source error. No runtime improvement has been measured and no new interacting coefficient has been established.

Extensions with explicit missing hypotheses. Other positive finite-range dispersions would require an appropriate global coordinate or partition, nondegenerate minima, controlled Jacobians and a summable heat remainder. Multiple minima change the leading prefactor. A field-dependent interacting refinement needs uniform comparison estimates on the desired field scale, not merely the ideal fugacity bound. An anisotropic spin theorem additionally needs the source's local geometry, path and capacity inputs for those rates. To identify a physical third coefficient one would first need an interacting comparison substantially sharper than $o(t^{-5/2})$, including the actual interaction terms at that order.

9 Evidence and conclusion

The literal read ledger records PRE hashes saved before display, all actual line intervals, and matching POST hashes. All 2994 lines of the 14 public text files and 819 lines of the two original private chapter bodies were read. Included private appendices and external companion proofs remain outside this scope. Captured-tree byte identity is provenance, not live-branch or theorem validation.

The owned executable checks separately exercise exact heat coefficients, product coefficients, rational nonreversible hitting identities, restricted energy norms and slot/site conversion. High-precision heat and tail evaluations are explicitly numerical diagnostics, not interval-certified proofs. The analytic theorems above provide the uniform bounds; finite fixtures do not validate the complete spin-system argument. Parent or other reports' checks are not included in the owned totals.

The contribution is a source-specific quantitative refinement of the ideal-evaluation and physical-capacity interfaces. It is not a claim of historical priority for lattice coefficients, Gaussian moments, energy duality or capacity identities. The corrected normalization sentence does not overturn the source's consistent final site formula. The global magnetization comparison, its geometric probability premises and the order of limits remain explicit.

References

[1] *The first lattice correction to Bloch's law*, cached source manuscript dated October 5, 2026, public repository OpenAI math, pinned tree `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Scientific text read locally in full; global theorem treated as imported.

[2] NIST Digital Library of Mathematical Functions, 10.32.1 and 10.40, modified Bessel integral and large-argument expansions. Displayed formulas and their domains checked; underlying monograph proofs not independently audited.

[3] M. Correggi, A. Giuliani and R. Seiringer, *Validity of the spin-wave approximation for the free energy of the Heisenberg ferromagnet*, Commun. Math. Phys. **339** (2015), 279–307, arXiv:1312.7873v2 (August 10, 2019). Abstract and version record checked, not the full proof.

[4] C. P. Hofmann, *Cubic Ideal Ferromagnets at Low Temperature and Weak Magnetic Field*, arXiv:1510.08930v2 (November 10, 2015). Abstract and version record checked; effective-theory scope retained.

[5] A. Gaudillière and C. Landim, *A Dirichlet principle for non reversible Markov chains and some recurrence theorems*, arXiv:1111.2445 (November 10, 2011). Abstract and metadata checked; classical context, not an imported substitute for the local proof given here.

[6] M. Subbarao, *Local moments*, RamCalBooks, volume 1, revision 2, original chapter `ch03-local-moments-rev2.tex`, confidential local corpus. Body lines 1–446 freshly read; subordinate included files not read.

[7] M. Subbarao, *Ram–CS*, RamCalBooks, volume 1, revision 2, original chapter `ch07-ramcs-rev2.tex`, confidential local corpus. Body lines 1–373 freshly read; subordinate included files not read.